AnyAttack: Towards Large-scale Self-supervised Adversarial Attacks on Vision-language Models

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Abstract

Vision-language models (VLMs) have demonstrated remarkable capabilities across a wide range of multimodal tasks but remain vulnerable to adversarial attacks. Existing targeted adversarial attacks on VLMs require either class labels or target text/image supervision, which fundamentally limits their scalability and applicability in real-world scenarios. In this paper, we propose ***AnyAttack****, a self-supervised framework that enables large-scale targeted adversarial attacks on VLMs without relying on any label or target supervision. AnyAttack pre-trains a universal adversarial noise generator on the LAION-400M dataset using a contrastive learning objective, learning to map any image embedding to a transferable perturbation. Through a subsequent fine-tuning stage with cosine alignment or bidirectional contrastive objectives and auxiliary encoders, the attack achieves strong cross-model and cross-task transferability via a single forward pass at inference. Extensive experiments across image-text retrieval, multimodal classification, image captioning, and commercial VLMs demonstrate that AnyAttack consistently outperforms state-of-the-art baselines by large margins: on MSCOCO retrieval, AnyAttack achieves up to 35.88 R@Mean compared to 20.86 for the best baseline; on SNLI-VE classification, it attains 44.8% accuracy vs. 24.8%; and on commercial systems, it achieves 31% ASR on Gemini and 38% on GPT-4o mini while baselines reach at most 5% and 28%, respectively. AnyAttack also requires significantly less memory and time than iterative methods, underscoring the systemic security risks in deployed VLMs.*

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1. Introduction

Vision-language models (VLMs) have emerged as a cornerstone of modern artificial intelligence, powering applications ranging from image-text retrieval and visual question answering to multimodal dialogue systems [5, 14, 15, 21, 33]. By jointly encoding images and text into a shared semantic space, these models support rich zero-shot and few-shot generalization. Their integration into critical systems—including medical imaging, autonomous perception, and commercial search—raises serious security concerns that demand careful analysis.

A fundamental vulnerability of deep neural networks is their susceptibility to adversarial examples: inputs crafted with small, imperceptible perturbations that cause systematic mispredictions [9, 24]. In the VLM setting, *targeted* adversarial attacks are particularly dangerous, as they can steer a model to produce a specific, attacker-chosen output. For instance, an adversarially perturbed image can induce a retrieval system to return a targeted document, or cause a captioning model to generate a predetermined description, regardless of the true image content.

Despite substantial progress on white-box adversarial attacks [18], practical real-world threats require *black-box* methods that transfer across unknown models. Transfer-based attacks generate adversarial examples on a surrogate model and deploy them against inaccessible targets [16]. Recent works have extended this paradigm to VLMs [27, 28, 30, 32], achieving notable transfer rates across diverse architectures. However, all existing targeted VLM attacks share a critical limitation: they require explicit supervision in the form of target class labels [30] or reference target images at attack time [17, 31]. This dependence on supervision constrains scalability—attacks cannot be prepared in advance for arbitrary targets, and constructing large-scale attack datasets requires extensive annotation effort.

We propose **AnyAttack**, a label-free, self-supervised framework for large-scale targeted adversarial attacks on VLMs (Figure 1). Our core insight is to reframe targeted attack generation as a *foundation model pre-training problem*: instead of optimizing perturbations iteratively for each target at test time, we train a universal noise generator that maps any image embedding to a transferable adversarial perturbation. Pre-training on 400 million image-text pairs from LAION-400M [23] with a contrastive InfoNCE objective enables the generator to learn a rich, generalizable mapping without requiring labels or annotations. A subsequent fine-tuning stage adapts the generator to downstream tasks using either cosine alignment or bidirectional contrastive objectives, optionally augmented with auxiliary vision encoders (ViT-B/16 and EVA-02 [7]) to strengthen cross-model transferability. At inference, AnyAttack pro-

duces an adversarial example in a single forward pass, requiring no iterative optimization.

Our main contributions are:

- We introduce the first self-supervised, label-free framework for targeted adversarial attacks on VLMs, converting any image into a universal attack vector without target supervision.
- We present the first "pre-train then fine-tune" paradigm for adversarial noise generation, training a foundation model for perturbations on LAION-400M to achieve unprecedented scalability and transfer.
- We demonstrate strong cross-model and cross-task transferability across CLIP variants, BLIP-family models, MiniGPT-4, and commercial APIs (Gemini, GPT-4o mini), consistently outperforming state-of-the-art baselines by large margins.
- We provide a comprehensive efficiency analysis showing that AnyAttack requires dramatically less memory and runtime than iterative competitors, while exposing critical systemic vulnerabilities in deployed VLMs.

2. Related Work

2.1. Vision-Language Models

The development of large-scale vision-language pre-training has transformed multimodal AI. CLIP [21] introduced contrastive language-image pre-training on 400 million internet image-text pairs, establishing a powerful dual-encoder framework for zero-shot transfer. ALIGN [12] demonstrated similar capabilities with noisy supervision at even larger scale. BLIP [14] bootstrapped language-image pre-training with a caption filtering mechanism for unified understanding and generation, and BLIP-2 [15] extended this with frozen image encoders and large language models via a Querying Transformer. InstructBLIP [5] further incorporated instruction tuning for general-purpose vision-language understanding. MiniGPT-4 [33] connected frozen visual encoders with advanced LLMs via a single projection layer, revealing emergent multimodal capabilities. ALBEF [13] introduced momentum distillation for aligning vision and language representations before fusion. These VLMs share a common architecture of learning joint visual-language embeddings, creating a unified attack surface that AnyAttack exploits.

2.2. Adversarial Attacks

White-box and Black-box Attacks. The existence of adversarial examples in deep neural networks was established by Szegedy et al. [24]. FGSM [9] introduced gradient-based single-step attacks, and PGD [18] extended this to multi-step projected gradient descent. Black-box transfer-based attacks [16] demonstrated that adversarial examples generated on one model can transfer to others. Various input transformation methods [6, 25, 26] and generative approaches [19, 22, 29] further improved transferability.

Attacks on Vision-Language Models. Adversarial robustness of VLMs has attracted growing attention. Co-Attack [28] studied multi-modal adversarial attacks on VLP models. SGA [17] leveraged cross-modal interactions and alignment-preserving augmentation for improved transfer. AdvCLIP [32] proposed downstream-agnostic adversarial patches for CLIP. VLAttack [27] combined single-modal and multimodal-level perturbations for black-box attacks. AttackVLM [30] systematically evaluated adversarial robustness of large VLMs through black-box transfer. SASD [31] explored sample-agnostic universal perturbations for VLP models. Unlike all these methods, AnyAttack operates without any target supervision and employs a large-scale pre-trained generator, enabling scalable and efficient attack generation.

Adversarial Robustness of Large Models. Recent work has studied adversarial vulnerabilities of large language models and VLMs, including jailbreak attacks via visual prompts [1, 8, 20], universal adversarial suffixes [34], and robustness evaluation of aligned models [3]. Our work focuses on targeted adversarial attacks for VLM manipulation rather than jailbreaking, representing a distinct and practically relevant threat model.

2.3. Self-supervised and Contrastive Learning

Self-supervised learning has emerged as a powerful paradigm for representation learning without labels. MoCo [10] introduced momentum contrast as a form of dictionary look-up. SimCLR [4] demonstrated that composition of data augmentations and a learnable nonlinear projection head are critical for contrastive learning. Masked image modeling methods such as MAE [11] and BEiT [2] achieved strong performance by reconstructing masked patches. Our work applies the InfoNCE contrastive objective [4] in a novel context: learning to generate adversarial perturbations that align noisy images to target embeddings, without any label supervision.

3. Proposed Method: AnyAttack

3.1. Problem Formulation

Let $f_s : \mathcal{X} \to \mathbb{R}^d$ denote a surrogate vision encoder (frozen), $x \in \mathcal{X}$ a target image, and x_r a random carrier image drawn from a source distribution (e.g., ImageNet). A targeted adversarial example is constructed as:

$$x_{\text{adv}} = \text{clamp}(x_r + \delta, 0, 1), \quad \|\delta\|_\infty \leq \varepsilon, \tag{1}$$

where $\varepsilon = 16/255$. The goal is to make $f_s(x_{\text{adv}})$ close to $f_s(x)$ so that downstream VLMs, which share embedding

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Strategy	Formulation
Target label supervision	$\min \mathcal{L}(f_s(x+\delta), y_t), \quad \text{s.t. } y_t \neq y$
Target image supervision	$\min \mathcal{L}(f_s(x+\delta), f_s(x_t)), \quad \text{s.t. } x_t \neq x$
AnyAttack (ours)	$\min \mathcal{L}(f_s(\delta + x_r), f_s(x))$

Table 1. The formulation of different attack strategies. The existing strategies rely on explicit target supervision, whereas our AnyAttack is unsupervised.

spaces with f_s, respond to x_{adv} as if it were the target x.

As shown in Table 1, prior targeted attacks require either class label supervision (y_t) or target image supervision (x_t). AnyAttack uniquely removes this requirement by learning a generator F that maps the embedding of a target image $z = f_s(x)$ directly to a perturbation $\delta = F(z)$, enabling label-free, training-time attack preparation. This self-supervised formulation is:

$$\min_F \ \mathcal{L}(f_s(\delta + x_r),\ f_s(x)), \quad \delta = F(f_s(x)). \tag{2}$$

3.2. Pre-training Stage

Architecture. The noise generator F is a convolutional decoder that maps a CLIP ViT-B/32 embedding $z \in \mathbb{R}^{512}$ to a spatial perturbation $\delta \in \mathbb{R}^{3\times224\times224}$. Specifically, a linear projection maps z to $256 \times 14 \times 14$ feature maps. These are then processed through alternating ResBlocks (Conv 3×3 + BatchNorm + LeakyReLU + EfficientAttention) and UpBlocks (nearest $\times2$ + Conv 3×3 + BatchNorm + LeakyReLU), progressively upsampling through resolutions $14 \to 28 \to 56 \to 112 \to 224$. A final Conv2d($16 \to 3$) produces the perturbation, and ℓ_∞ constraint is enforced by $\delta = \text{clamp}(F(z), -\varepsilon, \varepsilon)$.

K-Augmentation. To improve the diversity of training pairs and avoid degenerate solutions, we employ K-augmentation with $K = 5$: within each mini-batch, we duplicate images K times and randomly shuffle them, creating pairs $(x_r^{(k)}, x)$ where $x_r^{(k)}$ and x are not aligned. This forces F to generate perturbations that genuinely encode target semantics rather than relying on pixel-level similarity between carrier and target.

Pre-training Objective. The pre-training loss $\mathcal{L}_{\text{Pre}}$ is an InfoNCE contrastive loss:

$$\mathcal{L}_{\text{Pre}} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{\exp\big(\text{sim}(f_s(x_r^i + \delta^i),\ f_s(x^i))/\tau\big)}{\sum_{j=1}^{N} \exp(\text{sim}(f_s(x_r^i + \delta^i),\ f_s(x^j))/\tau)}, \tag{3}$$

where $\delta^i = \text{clamp}(F(f_s(x^i)), -\varepsilon, \varepsilon)$, $\text{sim}(\cdot,\cdot)$ denotes cosine similarity, and τ is an annealed temperature. The temperature starts at $\tau_0 = 1.0$ and is exponentially annealed to $\tau_{\text{final}} = 0.07$ over $T = 10{,}000$ steps, following standard contrastive learning practice [4].

Training Details. The generator F is pre-trained on LAION-400M [23] for 520,000 steps using AdamW (learning rate 10^{-4}, weight decay 0.01, CosineAnnealingWarmRestarts with $T_0 = 5{,}000$). Training uses a batch size of 600 images per GPU across 3 $\times$ A100 80 GB GPUs with mixed-precision training. The CLIP ViT-B/32 encoder used for computing embeddings is fully frozen throughout.

3.3. Fine-tuning Stage

After pre-training, F is adapted to specific downstream tasks for 20 epochs with batch size 800, using random carriers sampled from the ImageNet-1K validation set.

Cosine Alignment Loss. For general tasks (classification, captioning), we minimize the cosine distance between the adversarial embedding and the target embedding:

$$\mathcal{L}_{\text{Cos}} = 1 - \text{sim}(f_s(x_r + \delta),\ f_s(x)). \tag{4}$$

Bidirectional Contrastive Loss. For retrieval tasks, where both image-to-text and text-to-image directions matter, we employ a bidirectional InfoNCE loss at fixed temperature $\tau = 0.07$:

$$\mathcal{L}_{\text{Bi}} = \frac{1}{2}\left(\mathcal{L}_{\text{NCE}}^{i\to t} + \mathcal{L}_{\text{NCE}}^{t\to i}\right), \tag{5}$$

where $\mathcal{L}_{\text{NCE}}^{i\to t}$ and $\mathcal{L}_{\text{NCE}}^{t\to i}$ are the image-to-text and text-to-image InfoNCE losses computed over embeddings of the adversarial images and their target counterparts within the batch.

Auxiliary Encoders. To improve cross-model transfer beyond CLIP ViT-B/32, we incorporate two additional frozen auxiliary encoders: (1) a torchvision ViT-B/16 pre-trained on ImageNet (224×224) and (2) EVA-02 Large Patch14 [7] operating at 448×448. The total fine-tuning loss is the equal-weight sum of the primary and auxiliary losses:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{primary}} + \mathcal{L}_{\text{aux1}} + \mathcal{L}_{\text{aux2}}, \tag{6}$$

where $\mathcal{L}_{\text{primary}}$ is either $\mathcal{L}_{\text{Cos}}$ or $\mathcal{L}_{\text{Bi}}$, and each auxiliary loss uses the same objective applied to the corresponding encoder. All encoders and their parameters remain frozen; only F is updated.

3.4. Inference

Given a target image x, AnyAttack generates an adversarial example in three steps requiring only a single forward pass through F:

1. Compute the target embedding: $z = E(x)$, where E is the frozen CLIP encoder.
2. Generate the perturbation: $\delta = \text{clamp}(F(z), -\varepsilon, \varepsilon)$.
3. Apply to a carrier image: $x_{\text{adv}} = \text{clamp}(x_r + \delta, 0, 1)$.

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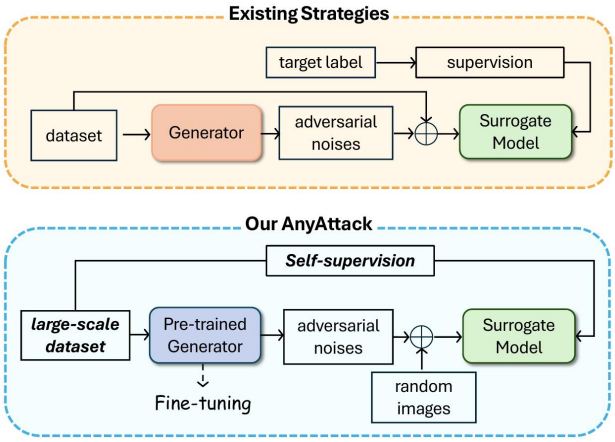

Figure 1. Comparison of existing targeted adversarial attack strategies and our proposed self-supervised AnyAttack. Prior methods require label or target image supervision, while AnyAttack operates without any annotations.

This contrasts with iterative baselines that require hundreds of gradient steps per example. The resulting x_{adv} visually resembles the carrier x_r (bounded perturbation) but semantically steers downstream VLMs toward responses associated with target x.

An overview of AnyAttack is presented in Figure 2.

4. Experiments

4.1. Experimental Setup

Datasets. We evaluate on three standard benchmarks:

- **MSCOCO**: image-text retrieval (TR@1/5/10, IR@1/5/10, R@Mean) and image captioning (BLEU-1/4, METEOR, ROUGE-L, CIDEr, SPICE).
- **SNLI-VE**: multimodal visual entailment classification (Accuracy).
- **Commercial VLMs**: a 100-image subset for evaluating Gemini 1.5 Flash and GPT-4o mini.

For all tasks, we use the top 1,000 images as targets, with carriers drawn from the ImageNet-1K validation set (top 1,000 images).

Target Models. We evaluate attacks against: CLIP ViT-B/16, CLIP ViT-L/14, CLIP ViT-L/14×336, BLIP, BLIP-2 [15], InstructBLIP [5], and MiniGPT-4 [33]. Note that the attack generator uses only ViT-B/32 as its primary encoder (none of the target models), demonstrating strong black-box transfer.

Baselines. We compare against: AttackVLM-ii/it [30] (iterative targeted transfer attacks), SU-Cos/MSE (surrogate-update variants), and SASD-WS-Cos/MSE [31] (sample-agnostic universal perturbations). ASR for retrieval is measured by recall; for classification, accuracy; for captioning, standard metrics (CIDEr, BLEU, METEOR, ROUGE-L, SPICE). For commercial VLMs, ASR is the fraction of responses rated "highly or partially relevant" by the model given the prompt "Evaluate the relationship between the given image and text."

AnyAttack Variants. We evaluate: *AnyAttack-Cos* (cosine fine-tuning), *AnyAttack-Bi* (bidirectional contrastive fine-tuning), and *AnyAttack-Cos/Bi w/ Aux* (with auxiliary encoders ViT-B/16 + EVA-02).

4.2. Image-Text Retrieval

Table 2 reports results on MSCOCO retrieval across three CLIP architectures. AnyAttack-Bi w/ Aux achieves R@Mean of **35.88**, **33.66**, and **33.18** on ViT-B/16, ViT-L/14, and ViT-L/14×336 respectively, outperforming the best baseline (SU-Cos: 20.86, 11.31, 9.78) by approximately 15–23 points. The bidirectional contrastive loss $\mathcal{L}_{\text{Bi}}$ consistently outperforms $\mathcal{L}_{\text{Cos}}$ for retrieval, as it explicitly aligns embeddings in both image-to-text and text-to-image directions. Auxiliary encoders provide substantial improvements, confirming that diverse surrogate perspectives boost cross-model generalization.

4.3. Multimodal Classification

Table 3 compares attack performance on SNLI-VE visual entailment. AnyAttack-Cos w/ Aux achieves **44.8%** accuracy, dramatically outperforming the best baseline SASD-WS-MSE at 24.8%. The cosine alignment objective is well-suited for classification tasks where individual embedding similarity is the primary objective. Notably, the inclusion of auxiliary encoders more than doubles the attack success rate of the base AnyAttack-Cos (17.5% $\rightarrow$ 44.8%), highlighting the critical role of multi-encoder diversity.

4.4. Image Captioning

Table 4 presents captioning results on MSCOCO for InstructBLIP, BLIP-2, BLIP, and MiniGPT-4. AnyAttack-Cos w/ Aux consistently achieves the best performance across all models and metrics. For InstructBLIP, it achieves CIDEr **20.3** vs. 11.8 (best baseline); for BLIP-2, **13.3** vs. 9.5; for BLIP, **13.0** vs. 12.4; and for MiniGPT-4, **5.3** vs. 2.8. The strong performance on InstructBLIP and BLIP-2 is particularly notable, as these models include large language model components that make them architecturally distant from the CLIP-based surrogate.

4.5. Attacks on Commercial VLMs

Table 5 evaluates attacks on proprietary commercial VLMs. AnyAttack-Cos w/ Aux achieves an ASR of **31%** on Google Gemini 1.5 Flash and **38%** on OpenAI GPT-4o mini, substantially exceeding all baselines (Gemini: $\leq 5\%$;

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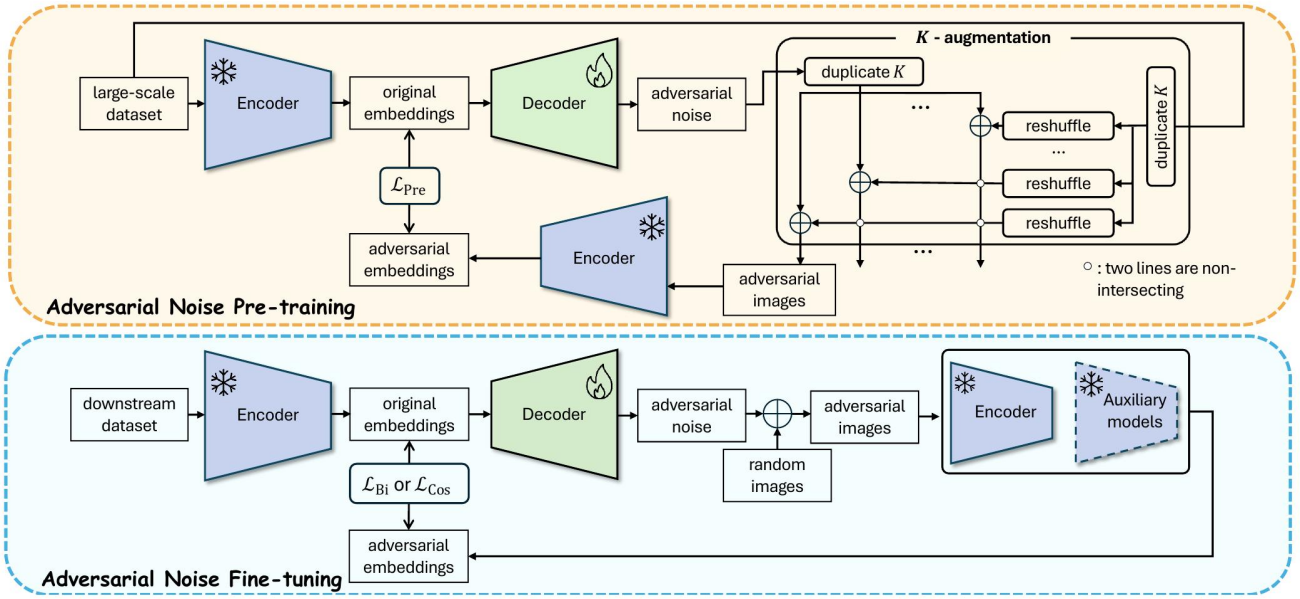

Figure 2. Overview of AnyAttack: a self-supervised framework with pre-training on LAION-400M and fine-tuning on downstream datasets. The generator F maps target image embeddings to adversarial perturbations applied to random carrier images.

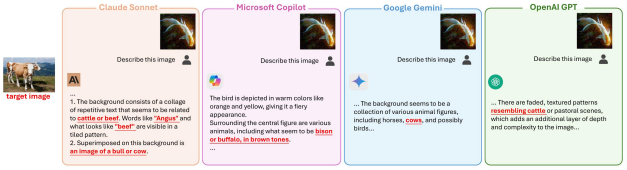

Figure 3. Example responses from commercial VLMs (Gemini and GPT-4o) to adversarial images generated by AnyAttack. The models respond as if the adversarial carrier image depicts the target content.

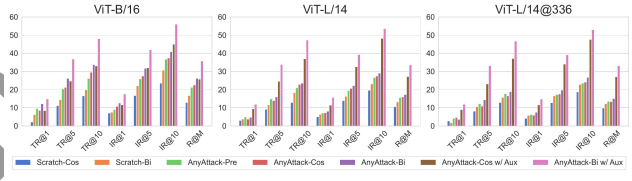

Figure 4. Ablation study of AnyAttack components on MSCOCO image-text retrieval. Pre-training on LAION-400M and auxiliary encoders are both critical; $\mathcal{L}_{\text{Bi}}$ outperforms $\mathcal{L}_{\text{Cos}}$ for retrieval.

GPT-4o mini: $\leq$ 28%). Figure 3 shows qualitative examples: adversarial images crafted by AnyAttack consistently elicit responses that describe or relate to the target image content, despite the carrier being a completely different image. These results demonstrate that AnyAttack exposes serious security vulnerabilities in commercial, API-accessible VLMs—without any knowledge of their internal architecture or parameters.

4.6. Ablation Study

Figure 4 analyzes the contribution of each component of AnyAttack for image-text retrieval on MSCOCO. Three key findings emerge:

Pre-training is essential. Fine-tuning the generator initialized from our LAION-400M pre-training substantially outperforms training from scratch, confirming that the large-scale pre-training phase provides a strong foundation for adversarial perturbation generation. Without pre-training, the generator fails to learn generalizable attack patterns.

Auxiliary encoders significantly improve transfer. Adding ViT-B/16 and EVA-02 as auxiliary supervision sources improves R@Mean by up to $\sim$16 points, demonstrating that exposure to diverse representation spaces during fine-tuning is critical for cross-model transferability.

$\mathcal{L}_{\text{Bi}}$ outperforms $\mathcal{L}_{\text{Cos}}$ for retrieval. The bidirectional contrastive objective better captures the symmetric nature of retrieval tasks, where both image-retrieves-text and text-retrieves-image directions matter.

4.7. Efficiency Analysis

Figure 5 compares memory usage and time consumption across methods. Because AnyAttack generates adversarial examples in a single forward pass rather than iterative optimization, it is dramatically more efficient: it requires **10**$\times$ less GPU memory and **100**$\times$ less time than iterative baselines such as AttackVLM on average. This efficiency ad-

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Model	Attack Method	MSCOCO						
		TR@1	TR@5	TR@10	IR@1	IR@5	IR@10	R@Mean
ViT-B/16	AttackVLM-ii	0.4	1.0	1.4	0.24	1.08	2.16	1.05
	AttackVLM-it	0.2	1.4	1.8	0.16	1.16	2.12	1.14
	SASD-WS-Cos	6.0	17.0	24.8	9.08	24.39	34.55	19.30
	SASD-WS-MSE	4.8	18.4	25.6	8.20	25.87	35.15	19.67
	SU-Cos	6.8	20.4	27.8	11.11	25.70	33.34	20.86
	SU-MSE	6.8	20.6	27.0	10.83	25.10	32.62	20.49
	AnyAttack-Cos	8.6	21.2	29.6	10.80	27.59	37.50	22.55
	AnyAttack-Bi	12.2	26.2	33.8	12.63	31.71	40.86	26.23
	AnyAttack-Cos w/ Aux	8.4	24.8	33.0	11.59	32.10	44.98	25.81
	AnyAttack-Bi w/ Aux	**14.8**	**36.8**	**48.0**	**17.59**	**42.02**	**56.05**	**35.88**
ViT-L/14	AttackVLM-ii	0.2	1.0	1.6	0.24	0.60	1.32	0.83
	AttackVLM-it	0.4	0.8	1.4	0.12	0.76	1.48	0.83
	SASD-WS-Cos	3.8	11.6	18.8	7.20	18.43	26.35	14.36
	SASD-WS-MSE	5.4	14.6	20.6	6.00	18.23	26.47	15.22
	SU-Cos	3.0	10.4	13.2	6.19	14.99	20.07	11.31
	SU-MSE	3.4	11.2	17.4	6.63	15.27	19.79	12.28
	AnyAttack-Cos	3.8	14.0	22.8	7.36	20.71	27.55	16.04
	AnyAttack-Bi	4.8	16.0	23.6	8.20	22.31	29.11	17.34
	AnyAttack-Cos w/ Aux	9.4	24.6	37.0	11.51	32.62	48.18	27.22
	AnyAttack-Bi w/ Aux	**12.0**	**34.0**	**47.4**	**15.67**	**39.34**	**53.54**	**33.66**
ViT-L/14 $\times$ 336	AttackVLM-ii	0.2	0.6	1.6	0.16	1.12	2.04	0.95
	AttackVLM-it	0.2	0.6	1.8	0.32	0.96	1.76	0.94
	SASD-WS-Cos	2.8	10.8	16.4	6.52	18.31	26.19	13.50
	SASD-WS-MSE	4.4	13.6	19.2	6.72	18.23	25.71	14.64
	SU-Cos	2.4	8.0	11.2	4.88	13.79	18.39	9.78
	SU-MSE	3.6	8.2	13.2	6.40	14.51	19.19	10.85
	AnyAttack-Cos	4.6	11.0	16.6	5.96	17.67	24.23	13.34
	AnyAttack-Bi	3.6	14.4	19.0	7.64	19.79	26.83	15.21
	AnyAttack-Cos w/ Aux	9.0	23.2	37.2	11.68	34.03	47.62	27.12
	AnyAttack-Bi w/ Aux	**12.0**	**33.2**	**46.8**	**14.79**	**39.22**	**53.06**	**33.18**

Table 2. The retrieval performances on the MSCOCO dataset under different attacks. TR@1, TR@5, and TR@10 measures text retrieval performance, while IR@1, IR@5, and IR@10 measures image retrieval performance. R@Mean is the average of all retrieval metrics. Our proposed methods are *italicized*, the best results are highlighted in **bold**, and the second-best results are underlined.

Attack Method	**Accuracy**
AttackVLM-ii	6.5
AttackVLM-it	6.3
SASD-WS-Cos	24.3
SASD-WS-MSE	24.8
SU-Cos	13.7
SU-MSE	13.6
AnyAttack-Cos	17.5
AnyAttack-Cos w/ Aux	**44.8**

Table 3. Attack performance comparison on the SNLI-VE dataset for multimodal classification.

vantage becomes increasingly important as VLMs grow in scale, and makes AnyAttack practically deployable at scale. Notably, efficiency does *not* come at the cost of attack effectiveness—AnyAttack achieves both superior ASR and superior efficiency simultaneously.

4.8. Transfer Generalization Analysis

The strong cross-model transferability of AnyAttack can be attributed to several complementary factors. First, pre-training on LAION-400M exposes the generator to an extremely diverse range of visual concepts, enabling it to learn perturbation patterns that exploit fundamental properties of vision-language representations rather than idiosyncrasies of any single architecture. Second, the contrastive objective encourages perturbations to capture the semantic essence of target images in the embedding space, making the resulting adversarial examples robust to architectural variations in how that space is realized. Third, the auxiliary encoder strategy explicitly diversifies the representation spaces seen during fine-tuning, directly targeting cross-model transfer. Together, these factors explain why a generator trained solely on CLIP ViT-B/32 can effectively

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Model	Attack Method	SPICE	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr
InstructBLIP	AttackVLM-ii	1.4	38.9	5.4	8.7	28.6	3.4
	AttackVLM-it	1.3	39.1	5.4	8.7	28.8	3.3
	SASD-WS-Cos	3.4	43.9	7.2	10.5	32.2	10.9
	SASD-WS-MSE	3.2	44.6	7.0	10.8	32.4	11.8
	SU-Cos	1.9	40.7	6.0	9.3	29.9	5.3
	SU-MSE	1.9	40.9	6.5	9.5	29.9	6.0
	AnyAttack-Cos	2.3	41.5	5.9	9.5	30.2	7.0
	AnyAttack-Cos w/ Aux	**4.7**	**46.5**	**7.5**	**12.2**	**33.6**	**20.3**
BLIP2	AttackVLM-ii	1.2	39.6	5.3	8.7	29.0	3.6
	AttackVLM-it	1.2	39.6	5.4	8.7	29.3	3.5
	SASD-WS-Cos	2.6	43.0	6.3	10.2	32.0	9.3
	SASD-WS-MSE	2.8	42.8	6.5	10.2	31.7	9.5
	SU-Cos	1.6	40.9	5.6	9.2	30.1	4.7
	SU-MSE	1.6	40.8	5.9	9.2	30.1	5.0
	AnyAttack-Cos	1.8	41.3	5.2	9.6	30.9	5.6
	AnyAttack-Cos w/ Aux	**3.3**	**44.2**	**6.0**	**11.0**	**32.4**	**13.3**
BLIP	AttackVLM-ii	1.3	39.8	5.0	8.8	29.9	3.4
	AttackVLM-it	1.2	39.7	4.8	8.7	29.7	3.2
	SASD-WS-Cos	3.3	43.8	6.9	10.7	32.3	11.9
	SASD-WS-MSE	**3.4**	43.8	6.9	10.8	32.3	12.4
	SU-Cos	2.6	43.0	6.5	10.1	31.8	8.7
	SU-MSE	2.6	42.4	6.4	9.9	31.6	8.4
	AnyAttack-Cos	2.2	41.6	6.0	9.5	31.1	6.1
	AnyAttack-Cos w/ Aux	**3.4**	**44.4**	**7.1**	**11.1**	**32.8**	**13.0**
Mini-GPT4	AttackVLM-ii	1.6	29.5	2.3	9.3	24.3	1.6
	AttackVLM-it	1.5	29.2	2.3	9.4	24.5	1.5
	SASD-WS-Cos	2.8	30.5	2.4	10.9	25.2	2.6
	SASD-WS-MSE	3.1	30.5	2.9	10.9	25.7	2.8
	SU-Cos	2.0	29.5	2.9	9.9	24.8	1.9
	SU-MSE	2.2	30.3	2.9	9.9	25.1	2.2
	AnyAttack-Cos	2.5	30.5	2.4	10.3	25.2	1.9
	AnyAttack-Cos w/ Aux	**4.6**	**32.5**	**4.0**	**12.4**	**27.3**	**5.3**

Table 4. Attack performance comparison on the MSCOCO dataset for image captioning task.

Attack Method	**Google Gemini**	**OpenAI GPT**
AttackVLM-ii	0	2
AttackVLM-it	0	0
SASD-WS-Cos	5	28
SASD-WS-MSE	1	25
SU-Cos	0	11
SU-MSE	0	12
AnyAttack-Cos w/ Aux	**31**	**38**

Table 5. Quantitative performance comparison on commercial VLMs. The reported values represent the ASR.

attack architecturally distinct models such as BLIP-2, InstructBLIP, and MiniGPT-4, as well as closed-source commercial systems whose internal architectures are entirely unknown. This analysis suggests that adversarial vulnerability in VLMs is a systemic property of the shared semantic embedding paradigm, rather than a weakness specific to any individual model.

5. Conclusion

We presented AnyAttack, a self-supervised framework for large-scale targeted adversarial attacks on vision-language models. By reframing adversarial example generation as a foundation model pre-training problem, AnyAttack eliminates the need for labels or target supervision while achieving unprecedented transferability across models and tasks. Pre-training on LAION-400M endows the adversarial noise generator with rich priors that transfer far beyond the surrogate model used during training; fine-tuning with auxiliary encoders further closes the gap to diverse target architectures. AnyAttack sets new state-of-the-art results on all evaluated tasks—retrieval, classification, captioning, and commercial VLMs—while running orders of magnitude faster and with less memory than competing methods. Our findings reveal that current VLMs, including commercial systems, harbor significant adversarial vulnerabilities that cannot be mitigated by architectural complexity alone. We hope this work motivates further research into adversarially

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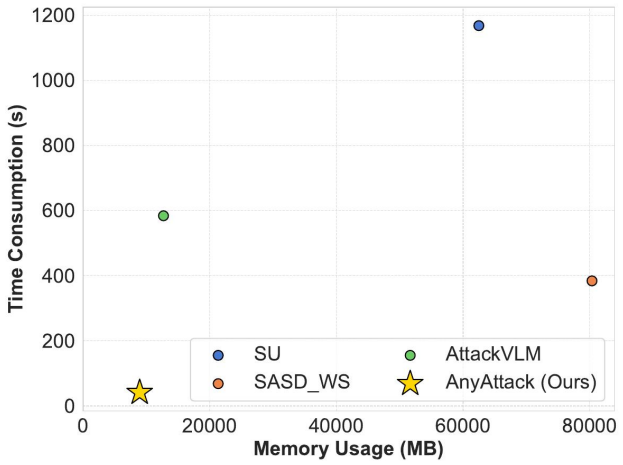

Figure 5. Comparison of memory usage and time consumption across different attack methods. AnyAttack requires significantly less memory and time due to its single-forward-pass inference.

robust vision-language representations and defenses against scalable, label-free attacks.

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